

Bar Jacobi

 barjacobi.com

 baraveh@gmail.com

 github.com/barjacobi

 [bar-jacobi](https://www.linkedin.com/in/bar-jacobi)

Education

University of Washington

2026 – Present

Ph.D. in Computer Science

Seattle, WA

Tel Aviv University

2024 – 2026

M.Sc. in Bioinformatics

Tel Aviv, Israel

Advisor: Prof. Adi Stern

Thesis: *Prediction of Viral Protein Function using Language Models*

Technion – Israel Institute of Technology

2017 – 2021

B.Sc. in Computer Science

Haifa, Israel

Research Experience

University of Washington

2026 – Present

Ruohola-Baker Lab, Department of Biochemistry

Seattle, WA

- Developing computational approaches to study transitions from healthy to aged cellular states using single-cell data and foundation models.
- Working closely with experimental biologists to connect computational predictions with experimental validation, including collaboration with the Institute for Protein Design.

Tel Aviv University

2024 – 2026

Stern Lab

Tel Aviv, Israel

Viral protein function prediction

- Developed machine-learning approaches for predicting viral protein function using protein language-model representations of sequence and structure.
- Designed family-aware evaluation strategies to measure generalization to evolutionarily distant viral proteins and quantify performance inflation under conventional similarity-based splits.
- Compared sequence- and structure-based representations and investigated how predictive performance changes with evolutionary distance and biological function.

Within-host evolution of SARS-CoV-2

- Studied intra-host viral diversity in chronically infected SARS-CoV-2 patients and acute-infection controls.
- Developed a replicate-concordance approach to distinguish genuine low-frequency variants from sequencing noise.
- Contributed to analyses of within-host evolution, resulting in a publication in *Virus Evolution*.

Technion – Israel Institute of Technology

2020 – 2021

Accelerated Computing Systems Lab

Haifa, Israel

Compilers for secure hardware enclaves

- Extended compiler-level memory sanitization to support secure enclave architectures.
- Implemented futex-based synchronization primitives for multithreaded runtime environments.

Publications

Ben Zvi, A., Rutsinsky, N., Jacobi, B., et al. (2025).

Diverse patterns of intra-host genetic diversity in chronically infected SARS-CoV-2 patients.

Virus Evolution. doi:10.1093/ve/veaf047

Industry Experience

Meta (Facebook)

2021 – 2024

Production Engineer, Facebook Lite

- Led infrastructure reliability work for Facebook Lite, serving more than 500 million monthly active users.
- Developed and maintained large-scale data pipelines, distributed monitoring systems, and reliability automation.

Teaching Experience

Tel Aviv University

2024 – 2025

Teaching Assistant, Python Programming for Engineers

Technion – Israel Institute of Technology

2021

Teaching Assistant, Advanced Operating Systems

she codes;

2016 – 2017, 2022

Volunteer Instructor, Java Programming

Fellowships & Awards

Excellence in Research Award, Tel Aviv University

2026

Faculty-wide annual research award to 30 outstanding graduate students in the Faculty of Life Sciences (500+ graduate students).

Warren Alpert Computational Biology and AI Network Fellowship

2025 – 2026

Selective fellowship in computational biology and AI through the Computational Genomics Summer Institute at UCLA.

Edmond J. Safra Center for Bioinformatics Fellowship

2024 – 2026

Graduate fellowship supporting training in bioinformatics at Tel Aviv University.

Dean's Excellence Award, Technion – Israel Institute of Technology

2019 – 2020

Semester-based academic distinction awarded to approximately the top 15% of Computer Science students (approximately 180–200 recipients per semester).

Presentations & Conferences

Edmond J. Safra Center for Bioinformatics Annual Retreat

Tel Aviv, 2026

Poster presentation; second place in the poster competition.

Computational Genomics Summer Institute (CGSI)

UCLA, 2025

Trainee and presenter.

SBCR Conference: Viruses Across Domains of Life

Tel Aviv, 2025

Poster presentation.

Israeli Conference on Hardware and Side-Channel Attacks

2021

Competition finalist.

Leadership & Service

Alumni Speaker, Technion Career Week

2025

CV workshop for students.

Student Representative, Technion Student Association

2018 – 2020

Technical Skills

Programming	Python, Bash, C++
Machine Learning	PyTorch, scikit-learn, protein language models
Bioinformatics	NGS analysis, phylogenetics, MMseqs2, iVar
Data Analysis	pandas, NumPy, SQL, scientific visualization
Computing	Linux, Docker, Git, HPC environments